

TECHNICAL REFERENCE GUIDE

Satellite Remote Sensing Sensors & Applications

With Bhopal & Sehore Region Water Bodies Case Study

Sentinel-2 MSI	NDWI, NDVI, Water Body Mapping
Landsat-8/9 OLI	Historical & Multispectral Analysis
Landsat-8/9 TIRS	Land Surface Temperature
Sentinel-1 SAR	Cloud-Independent Water Detection
Case Study	Bhopal & Sehore Water Bodies, Madhya Pradesh, India

July 2026

1. What is Satellite Remote Sensing?

Satellite remote sensing acquires information about Earth's surface using sensors on orbiting satellites. These sensors detect electromagnetic radiation - reflected sunlight, emitted thermal energy, or radar returns - and convert it into digital data for analysis.

Two types of sensors are used:

- **Passive Sensors:** Detect natural radiation (sunlight/thermal). Need clear skies. Examples: MSI, OLI, TIRS.
- **Active Sensors:** Emit their own signal and measure the return. Work through clouds, day and night. Example: SAR.

Electromagnetic Spectrum Regions

Region	Wavelength	Sensor	What It Measures
Visible	0.4-0.7 um	MSI, OLI	Surface color, chlorophyll
Near-Infrared	0.7-1.3 um	MSI, OLI	Vegetation health, water boundaries
Short-Wave IR	1.3-3.0 um	MSI, OLI	Soil moisture, minerals, snow
Thermal IR	8-14 um	TIRS	Surface temperature
Microwave (C)	3.8-7.5 cm	SAR	Surface roughness, floods

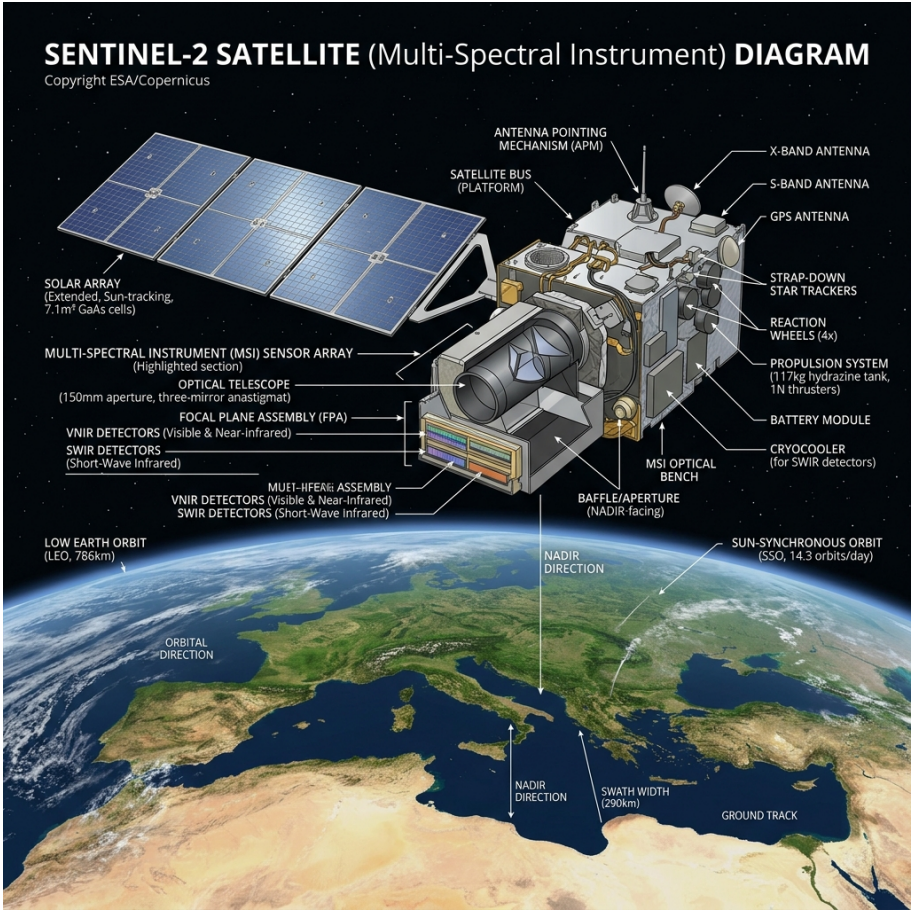
Key Sensor Parameters

Parameter	What It Means	Example
Spatial Resolution	Smallest area visible on ground	10 m (Sentinel-2)
Spectral Resolution	Number of color bands captured	13 bands (Sentinel-2)
Radiometric Resolution	Sensitivity to brightness levels	12-bit = 4096 levels
Temporal Resolution	How often the satellite revisits	5 days (Sentinel-2)

2. Sentinel-2 MSI - Multispectral Instrument

PRIMARY SENSOR

ESA Copernicus | Sentinel-2A (2015) & 2B (2017)



Sentinel-2 satellite with MSI sensor in orbit at 786 km

ALTITUDE	SWATH WIDTH
786 km	290 km
BANDS	REVISIT
13	5 Days
BEST RESOLUTION	BIT DEPTH
10 m	12-bit

How It Works

MSI uses a pushbroom scanner: a linear array of CMOS detectors captures one line at a time as the satellite moves forward. 13 bands from visible to short-wave infrared enable vegetation, water, and land cover analysis.

Spectral Bands

Band	Name	nm	Res	Use
B2	Blue	490	10 m	Water, soil
B3	Green	560	10 m	NDWI (water index)
B4	Red	665	10 m	NDVI (vegetation)
B8	NIR	842	10 m	NDVI, NDWI, biomass
B11	SWIR 1	1610	20 m	Moisture, snow/ice
B12	SWIR 2	2190	20 m	Minerals, geology

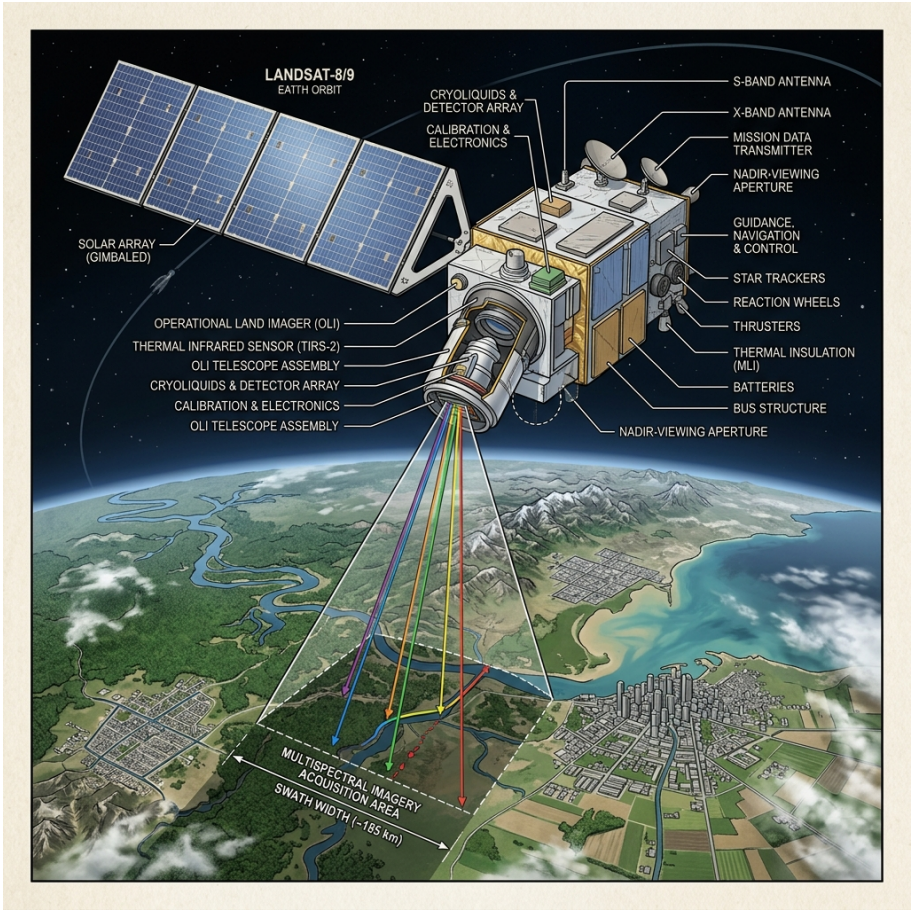
Main Applications

- **NDVI mapping:** Vegetation health using Red & NIR bands
- **NDWI mapping:** Water body delineation using Green & NIR bands
- **Water body mapping:** 10 m resolution lake/river identification
- **Agriculture:** Crop classification, drought monitoring

3. Landsat-8/9 OLI - Operational Land Imager

MULTISPECTRAL

NASA/USGS | Landsat-8 (2013) & Landsat-9 (2021)



Landsat-8/9 satellite with OLI and TIRS instruments

ALTITUDE 705 km	SWATH WIDTH 185 km
BANDS 9	REVISIT 16 Days (single)
RESOLUTION 30 m (15 m PAN)	BIT DEPTH 12/14-bit

How It Works

OLI is a pushbroom sensor with 9 spectral bands. Silicon detectors handle visible/NIR bands, while HgCdTe (Mercury Cadmium Telluride) detectors cover SWIR bands. The 50+ year Landsat archive (since 1972) makes it the most valuable dataset for historical change detection.

Spectral Bands

Band	Name	Wavelength	Res	Use
2	Blue	0.45-0.51 μm	30 m	Bathymetry, soil
3	Green	0.53-0.59 μm	30 m	Vegetation reflectance
4	Red	0.64-0.67 μm	30 m	Vegetation NDVI
5	NIR	0.85-0.88 μm	30 m	Biomass, shorelines
6	SWIR 1	1.57-1.65 μm	30 m	Moisture, geology

8	PAN	0.50-0.68 um	15 m	Sharpening images
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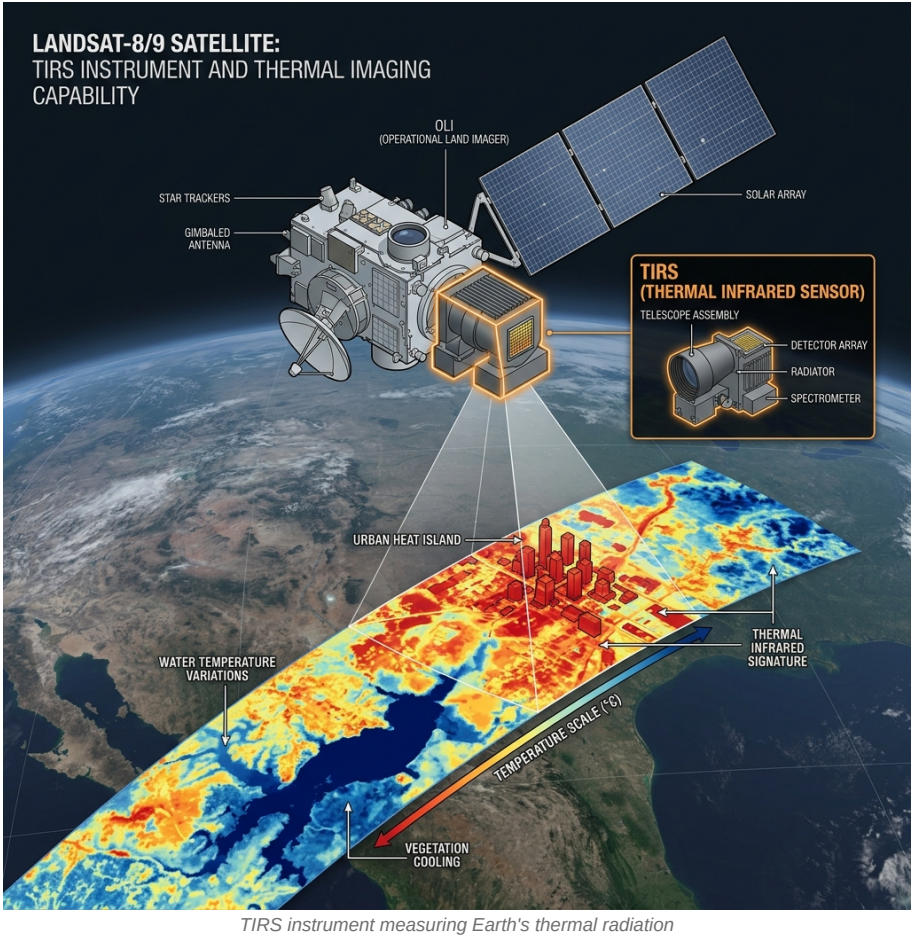
Main Applications

- **Historical analysis:** 50+ years of continuous Earth observation data
- **Land cover mapping:** 9-band classification of urban, forest, water, crops
- **NDVI/NDWI:** Vegetation and water index computation
- **Geology:** SWIR bands for mineral and soil mapping

4. Landsat-8/9 TIRS - Thermal Infrared Sensor

THERMAL

NASA Goddard | Rides on Landsat-8/9 satellite



BANDS 2 (Thermal)	RESOLUTION 100 m
ACCURACY +/- 1 K	DETECTOR QWIP (GaAs)
COOLING Cryocooler 43 K	SWATH 185 km

How It Works

TIRS measures thermal infrared radiation (10-12.5 μm) emitted by Earth's surface. It uses Quantum Well Infrared Photodetectors (QWIP) made from GaAs/AlGaAs semiconductors, cooled to 43 K to reduce noise. The two bands enable atmospheric correction using the split-window technique.

Band	Wavelength	Resolution	Purpose
Band 10	10.60-11.19 μm	100 m	Surface temperature measurement
Band 11	11.50-12.51 μm	100 m	Atmospheric correction

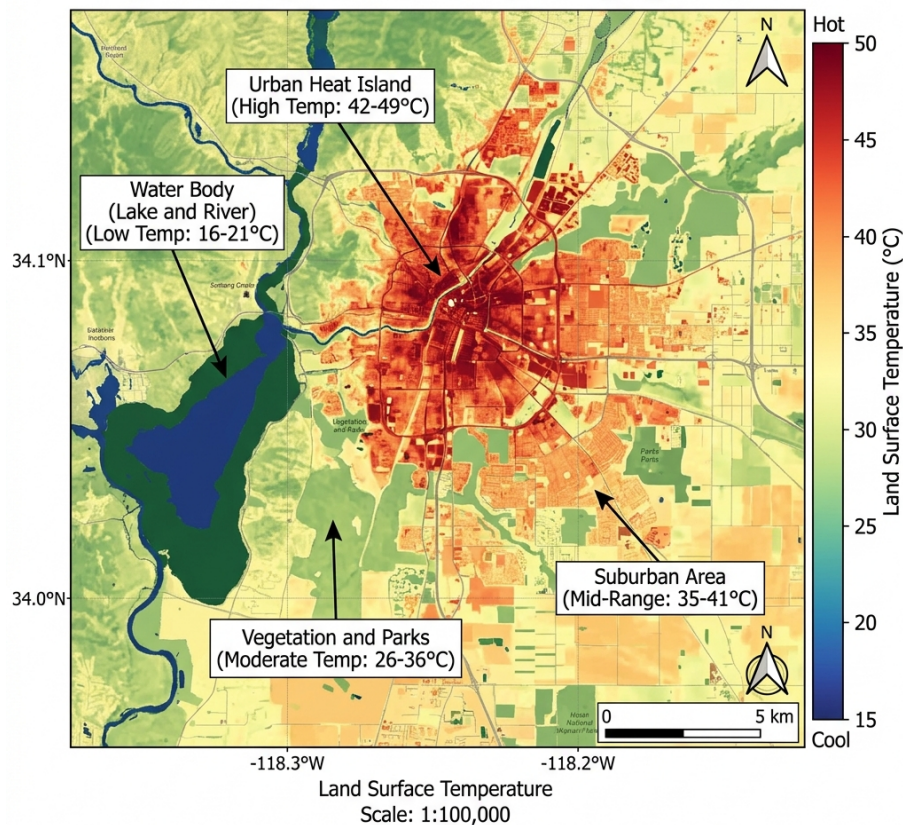
Temperature Retrieval (Simplified)

Raw satellite data (Digital Numbers) is converted to temperature in 3 steps:

- **Step 1:** DN -> Spectral Radiance (using calibration constants from metadata)
- **Step 2:** Radiance -> Brightness Temperature (inverting Planck's radiation law)

- **Step 3:** Brightness Temp -> LST (correcting for surface emissivity)

Satellite Land Surface Temperature (LST) Mapping Satellite Thermal Infrared (TIR) Data - August 14, 2023



Land Surface Temperature map showing urban heat island effect

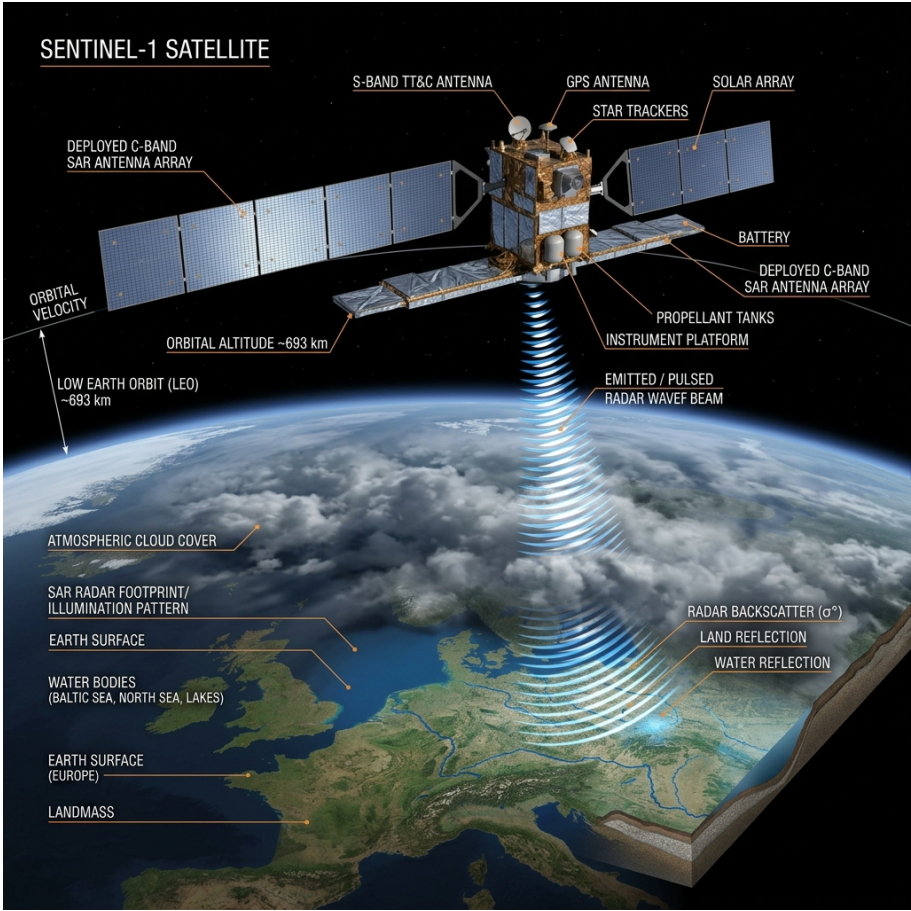
Main Applications

- **Surface temperature:** Mapping land/water temperature from space
- **Urban heat islands:** Cities are hotter than surrounding rural areas
- **Evapotranspiration:** Water loss from soil/plants for irrigation planning
- **Fire detection:** Identifying active wildfires and burn areas

5. Sentinel-1 SAR - Synthetic Aperture Radar

ACTIVE RADAR

ESA Copernicus | Sentinel-1A (2014) & 1C (2024)



Sentinel-1 with SAR antenna transmitting radar through clouds

FREQUENCY 5.405 GHz (C-band)	WAVELENGTH 5.547 cm
ANTENNA 12.3 m x 0.82 m	ALTITUDE 693 km
BEST RESOLUTION 5 m x 5 m	REVISIT 6 Days

How It Works

SAR is an active sensor - it sends microwave radar pulses and measures the echo. Unlike optical sensors, it works through clouds and at night. The 12.3 m phased array antenna with 280 transmit/receive modules creates a 'synthetic' aperture much larger than the physical antenna, achieving high resolution.

Mode	Swath	Resolution	Use
Stripmap	80 km	5 x 5 m	Emergencies, small areas
Interferometric Wide	250 km	5 x 20 m	Default land monitoring
Extra Wide	400 km	20 x 40 m	Maritime, sea ice

Why Water Appears Dark in SAR

Smooth water reflects radar away from the satellite (like a mirror), so very little signal returns. Land and buildings scatter radar back. This makes water bodies stand out as dark patches, enabling flood mapping even through thick clouds during monsoons.

Main Applications

- **Flood mapping:** Cloud-free water detection during monsoons and storms
- **Water body monitoring:** All-weather reservoir and lake tracking
- **Ground deformation:** mm-level subsidence detection using InSAR
- **Ship detection:** Maritime surveillance, oil spill monitoring

6. NDVI & NDWI - Spectral Indices

Spectral indices are simple math formulas using two spectral bands to highlight specific features. They output values from -1 to +1, making interpretation easy and consistent.

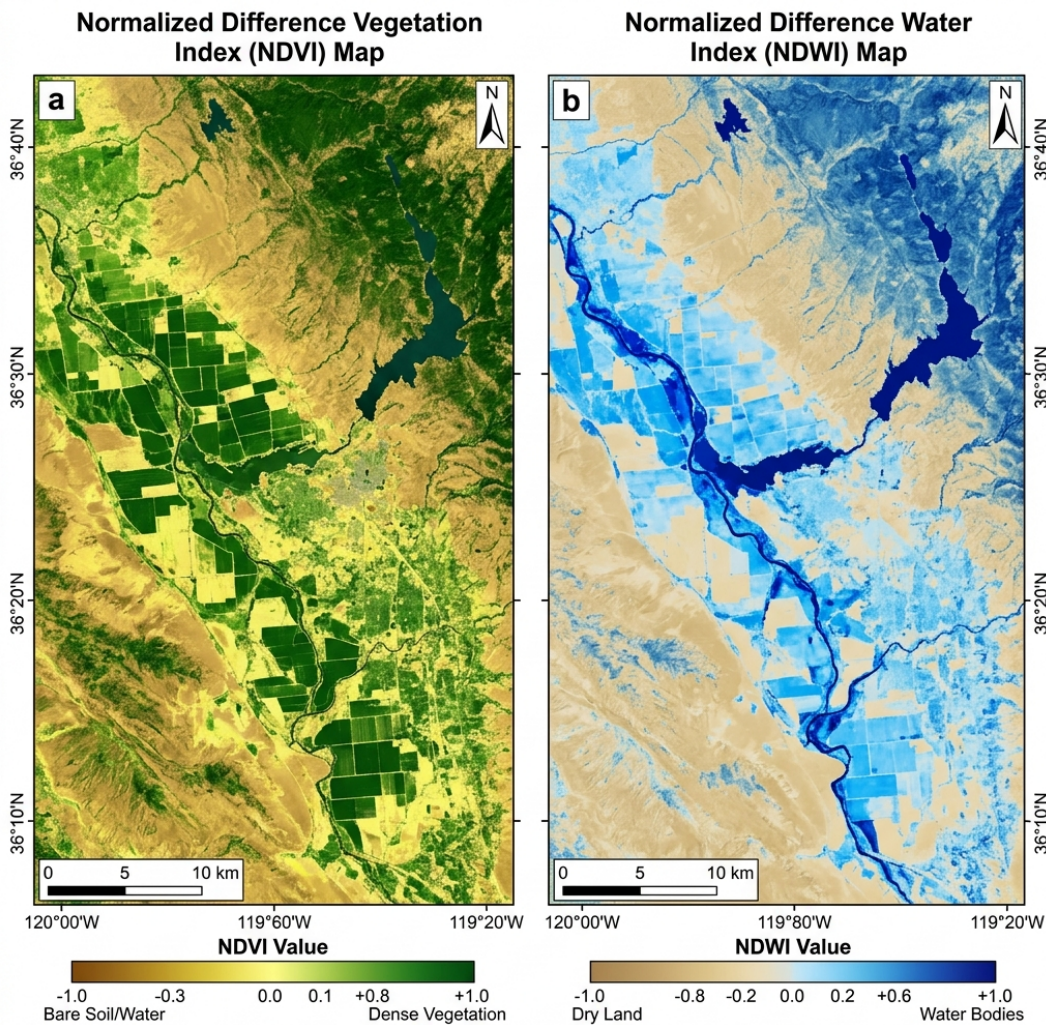


Figure 1: Side-by-side NDVI and NDWI comparison of Central California
NDVI (vegetation health) vs NDWI (water detection) comparison

NDVI - Vegetation Index

NDVI FORMULA

NDVI = (NIR - Red) / (NIR + Red)

Sentinel-2: (B8-B4)/(B8+B4) | Landsat: (B5-B4)/(B5+B4)

Why it works: Healthy plants absorb red light for photosynthesis but strongly reflect near-infrared. More vegetation = higher NDVI.

NDVI Value	Meaning	Surface Type
< 0	Not vegetation	Water, snow, clouds
0 to 0.1	Bare ground	Rock, sand, bare soil
0.1 to 0.3	Sparse vegetation	Grassland, stressed crops
0.3 to 0.6	Moderate vegetation	Cropland, mixed forest
0.6 to 1.0	Dense vegetation	Healthy forest, crops

NDWI - Water Index

NDWI FORMULA (MCFEETERS, 1996)

$$\text{NDWI} = (\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$$

Sentinel-2: (B3-B8)/(B3+B8) | Landsat: (B3-B5)/(B3+B5)

Why it works: Water absorbs NIR strongly but reflects green moderately. So NDWI is positive for water and negative for land/vegetation.

NDWI Value	Meaning	Surface Type
0.3 to 1.0	Open water	Lakes, rivers, reservoirs
0 to 0.3	Wet surfaces	Shallow water, flooded areas
< 0	Not water	Vegetation, soil, buildings

MODIFIED NDWI (MNDWI)

MNDWI = (Green - SWIR) / (Green + SWIR)

Better at separating water from buildings in urban areas

7. Sensor Comparison at a Glance

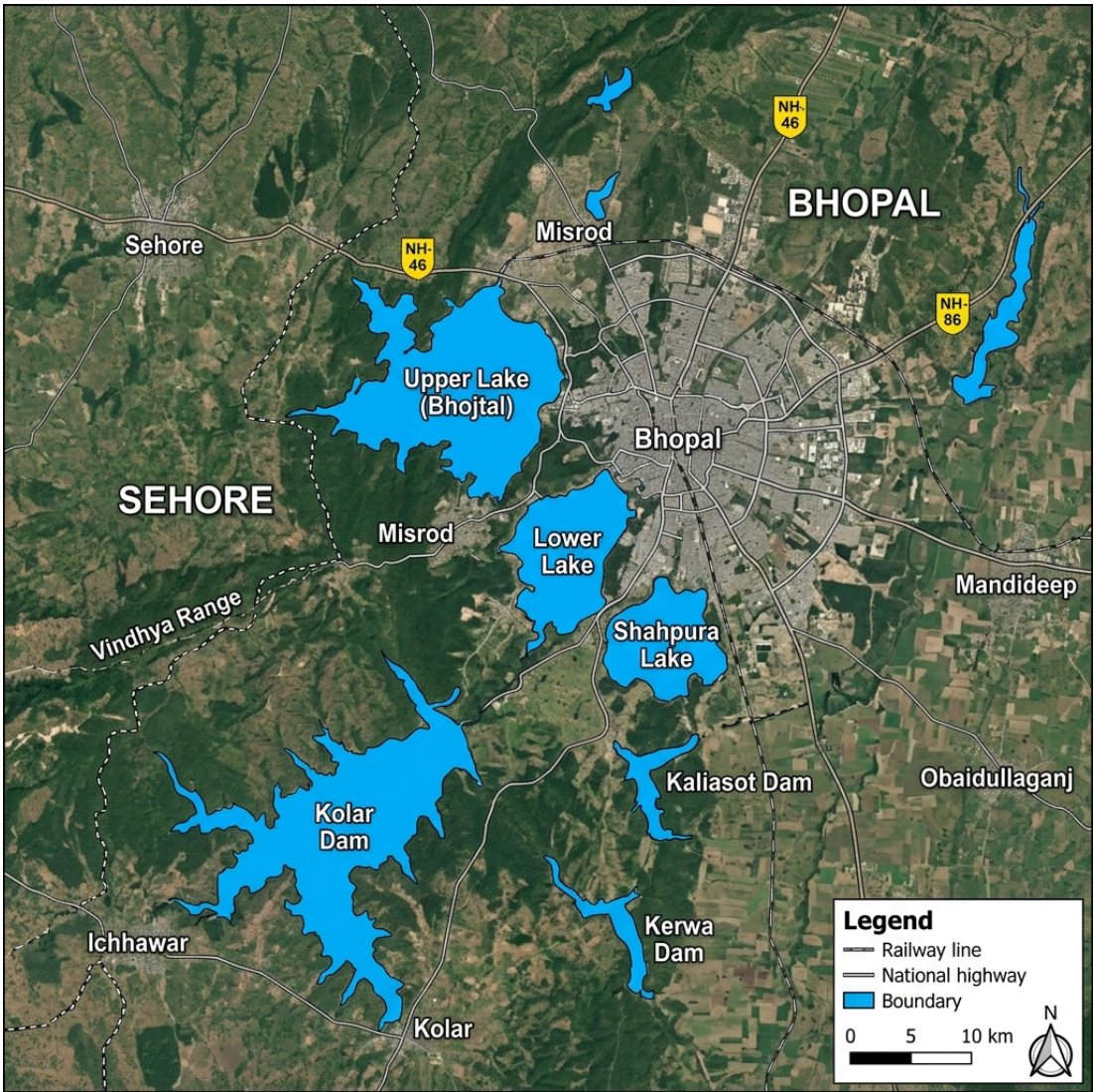
Feature	Sentinel-2 MSI	Landsat OLI	Landsat TIRS	Sentinel-1 SAR
Agency	ESA	NASA/USGS	NASA	ESA
Type	Passive Optical	Passive Optical	Passive Thermal	Active Radar
Bands	13	9	2	1 (dual-pol)
Best Res.	10 m	15 m (PAN)	100 m	5 m
Swath	290 km	185 km	185 km	250 km
Revisit	5 days	8 days (pair)	8 days	6 days
Works in clouds?	No	No	No	YES
Works at night?	No	No	Yes	YES
Main use	NDVI/NDWI	Historical data	Temperature	Flood mapping
Data cost	Free	Free	Free	Free

How to Choose the Right Sensor

- Need high-res vegetation/water mapping? -> Sentinel-2 MSI (10 m, 5-day revisit)
- Need historical trends (decades)? -> Landsat OLI (data since 1972)
- Need surface temperature? -> Landsat TIRS (thermal bands)
- Need cloud-free observation? -> Sentinel-1 SAR (works through clouds)
- Best approach: Combine all four for the most complete picture!

8. Bhopal & Sehore Water Bodies

Bhopal, the 'City of Lakes,' and adjacent Sehore district contain some of central India's most important water bodies. These serve as drinking water sources, irrigation systems, and ecological habitats. Satellite remote sensing is the most effective tool to monitor their health over time.



Bhopal & Sehore region showing major water bodies

Bhopal District - Major Water Bodies

Water Body	Location	Area	Key Use	Main Concern
Upper Lake (Bhojtal)	23.25N, 77.33E	~36 sq km	40% drinking water	May shrink 40% by 2050
Lower Lake	23.24N, 77.41E	~1.29 sq km	Recreation	Sewage, encroachment
Kaliasot Dam	23.20N, 77.41E	Reservoir	Drinking, flood ctrl	Siltation, sewage
Kerwa Dam	23.16N, 77.36E	Reservoir	Ecology, tourism	Illegal land filling
Shahpura Lake	23.20N, 77.42E	Stable	Urban recreation	Urban encroachment
Hathaikheda Lake	23.27N, 77.47E	Small lake	Water storage	Unregulated development

Upper Lake (Bhojtal / Bada Talab) - In Detail

The Upper Lake is Bhopal's largest and most critical water body. Built in the 11th century by Raja Bhoj, it is a Ramsar Wetland Site (since 2002) supplying ~40% of Bhopal's drinking water. Its surface area is approximately 36 sq km at full capacity.

Satellite Monitoring of Upper Lake

- Landsat time series (1984-present): Shows gradual decrease in water spread area
- Sentinel-2 NDWI at 10 m: Tracks seasonal water extent changes every 5 days
- MNDWI performs better than NDWI for this lake (research findings)
- Mann-Kendall test on Landsat data confirms significant decreasing trend
- AI models predict ~40% shrinkage by 2050 under climate change scenarios

Sehore District - Major Water Bodies

Kolar Dam - The Lifeline of Bhopal

Parameter	Detail
Location	Sehore District, 35 km SW of Bhopal, on Kolar River
Dam Height / Length	45 m (148 ft) / 590 m
Storage Capacity	265-270 million cubic meters (MCM)
Catchment Area	506-508 sq km
Water Supply	~153 MLD = ~60% of Bhopal's drinking water
Irrigation	36,000 - 60,000+ hectares (Sehore & Raisen)
Other Uses	Inland fisheries, nature tourism

Rivers in Sehore District

River	Basin	Significance
Kolar	Narmada (right bank)	Feeds Kolar Dam, primary drainage
Ajnal	Narmada	Proposed dam at Nasrullahganj
Newaj	Chambal/Yamuna	Major drainage, eastern Sehore
Paru	Narmada	Agricultural irrigation
Papnas / Parva / Sewan / Sip	Narmada	Minor drainage and irrigation

9. Applying Sensors to Bhopal-Sehore Region

Each sensor has a specific role in monitoring the Bhopal-Sehore water bodies:

Sensor	What It Does for This Region	Best For
Sentinel-2 MSI	10 m NDWI maps of all lakes	Seasonal water extent
Sentinel-2 MSI	NDVI for catchment vegetation health	Green belt monitoring
Landsat OLI	Water area trends since 1984	Decades of history
Landsat OLI	Land use change in catchments	Urbanization tracking
Landsat TIRS	Lake surface temperature	Thermal analysis
Landsat TIRS	Urban heat island mapping	Bhopal city thermal
Sentinel-1 SAR	Monsoon flood monitoring	Cloud-free mapping
Sentinel-1 SAR	Reservoir level estimation	Kolar Dam tracking

Environmental Challenges

- **Urbanization:** Unauthorized construction in catchment areas reduces water capacity. Sentinel-2 NDWI tracks this.
- **Siltation:** Sediment deposition makes lakes shallower over time. Multi-decadal Landsat data shows trends.
- **Climate change:** Upper Lake may shrink 40% by 2050. Landsat archive provides the baseline for AI predictions.
- **Pollution:** Sewage and runoff affect water quality. Sentinel-2 spectral data provides turbidity estimates.
- **Monsoon variability:** Sentinel-1 SAR maps floods even through heavy monsoon cloud cover.

Recommended Workflow for Monitoring

1. Download Sentinel-2 & Landsat data (free from Copernicus Hub / EarthExplorer)
2. Apply atmospheric correction (Sen2Cor for Sentinel-2)
3. Compute NDWI to identify water pixels (threshold > 0.0)
4. Use Sentinel-1 SAR for cloud-covered dates (backscatter thresholding)
5. Compare with historical Landsat imagery for change detection
6. Use TIRS for surface temperature analysis of reservoirs
7. Combine all layers for a complete water body assessment

Where to Get the Data (All Free)

- Sentinel-2 & Sentinel-1: Copernicus Open Access Hub (scihub.copernicus.eu)
- Landsat-8/9: USGS EarthExplorer (earthexplorer.usgs.gov)
- Indian water body data: Bhuvan-WBIS by NRSC/ISRO (bhuvan.nrsc.gov.in)
- Cloud processing: Google Earth Engine (code.earthengine.google.com)
- Desktop GIS: QGIS (free, open source) or ArcGIS

10. References

- [1] ESA (2015). Sentinel-2 User Handbook. sentinels.copernicus.eu
- [2] USGS (2019). Landsat 8-9 Calibration and Validation.
- [3] Reuter et al. (2015). "Thermal Infrared Sensor on Landsat 8." Remote Sensing, 7(1).
- [4] Torres et al. (2012). "GMES Sentinel-1 mission." Remote Sensing of Environment, 120.
- [5] McFeeters (1996). "NDWI for delineation of open water." Int J Remote Sensing, 17(7).
- [6] Rouse et al. (1974). "Monitoring vegetation with ERTS." NASA SP-351.
- [7] Xu (2006). "Modified NDWI to Enhance Open Water." Int J Remote Sensing, 27(14).
- [8] Richards (2022). Remote Sensing Digital Image Analysis, 6th ed. Springer.
- [9] CGWB. District Groundwater Reports - Bhopal & Sehore. Central Ground Water Board.
- [10] NRSC/ISRO. Bhuvan-WBIS Water Bodies Information System. bhuvan.nrsc.gov.in
- [11] JICA. Bhopal Lake Conservation Project Reports.

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